Empirical Methods for Thesis Writers

Causal Inference and Research Design

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How Not to Do Empirical Research

ASK WELL

Do I Need to Avoid Dark Chocolate Now?

A recent report found high concentrations of cadmium and lead in dark chocolate. There's no reason to panic, experts say, but it's worth a closer look.

ASK WELL

Is Chocolate Good for You?

Studies suggest that cocoa might benefit health, but it's unclear how that may translate to a typical bar of chocolate.

Why Chocolate May Be Good for the Heart

Share full article 69



Andrew Scrivani for The New York Times

How not to do empirical research (continued)

ECONOMIC POLICY

Want to live longer? Send your kids to college



Everyone is smiling, because everyone lives longer. (AP Photo/The Daily News-Record, Michael Reilly)

Parents of college graduates live longer.

- What is the mechanism?
- What else differs between families whose kids go to college and families whose kids don't?
- If you could run an experiment to test this, what would it look like?

Hold on to that last question. It is the most useful habit in this lecture.

What would Tolstoy say?

All good empirical papers are alike; each flawed paper is flawed in its own way.

A good empirical paper has

- A clear research question
- A plausible mechanism
- A well-defined counterfactual: what would have happened otherwise?

Running example for today: what is the effect of writing a senior thesis at Stern on your lifetime earnings?

- What are the mechanisms? What is the counterfactual?

Plan for today

1. The counterfactual: what a causal effect is, and why comparing groups usually fails
2. What to control for: drawing your assumptions as pictures
3. Where counterfactuals come from: a map of research designs
    - Lotteries and natural experiments (instrumental variables)
    - Before and after: event studies and difference-in-differences
    - One treated unit: synthetic control
    - Cutoffs: regression discontinuity
4. A checklist for your thesis

Goal: you leave knowing *which questions to ask*. The formulas are easy to look up later.

The Counterfactual

Defining the counterfactual

Each student i has two potential outcomes:

- $Y_i(1)$: lifetime earnings if i writes a thesis
- $Y_i(0)$: lifetime earnings if i does not

The causal effect for student i is $Y_i(1) - Y_i(0)$.

The fundamental problem of causal inference: we see each student in only one world.

Student	$Y_i(1)$	$Y_i(0)$	Wrote thesis?
Ana	\$4.1M	?	Yes
Ben	?	\$3.2M	No
Cal	\$3.6M	?	Yes

Every method today is a way of filling in the ?s with someone or something else.

The naive comparison

Compare average earnings of thesis writers to non-writers:

$$\underbrace{\mathbb{E}[Y \mid \text{thesis}] - \mathbb{E}[Y \mid \text{no thesis}]}_{\text{what we see}} = \underbrace{\text{effect of the thesis}}_{\text{what we want}} + \underbrace{\mathbb{E}[Y(0) \mid \text{thesis}] - \mathbb{E}[Y(0) \mid \text{no thesis}]}_{\text{selection bias}}$$

- Selection bias: would thesis writers have earned more even without the thesis?
- Almost certainly yes: higher GPA, more ambition, better recommendations, more interest in research.
- Same problem as chocolate and longevity. The groups were different before the "treatment."

Why experiments work

Suppose we flipped a coin to decide who writes a thesis.

- Then the two groups are the same on average, in everything: GPA, ambition, family background, things we can't measure.
- Selection bias is zero, and the simple comparison of means is the causal effect.

We usually can't flip the coin. So the rest of this lecture is about finding situations that are

as good as random

and being honest about what has to be true for that to hold.

Effect for whom?

"The effect" is not one number. Be clear about *whose* effect you estimate:

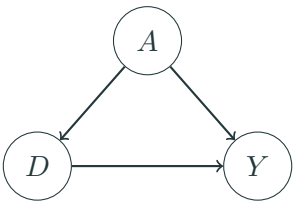
- Average effect (ATE): if we made a random Stern student write a thesis.
- Effect on the treated (ATT): for the students who actually chose to write one.
- Local effect (LATE): for students whose choice was changed by your source of variation (more on this soon).

These can differ a lot when people choose treatment because they expect to benefit.

- Your thesis should say which one you estimate, and whether it speaks to the policy question you care about (external validity).

What Should I Control For?

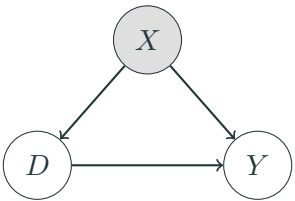
Drawing your assumptions



D = thesis, Y = earnings, A = ability / ambition

- A causal diagram (DAG): arrows are causal effects.
- Two routes connect D and Y:
  - $D \to Y$: the causal path we want
  - $D \leftarrow A \to Y$: a backdoor path, which creates correlation even if the thesis does nothing
- Every missing arrow is an assumption. Drawing the picture forces you to state them.
- Your research design is a plan to block every backdoor path.

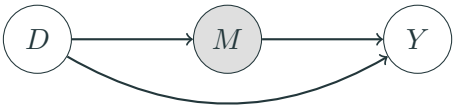
Good controls: blocking a backdoor



Shaded = controlled for. X = GPA.

- A confounder causes both treatment and outcome.
- Controlling for it (regression, matching) blocks that backdoor.
- If every backdoor runs through variables you observe, controls are enough. This is selection on observables.
- The catch: ambition, network, and family are hard to measure.

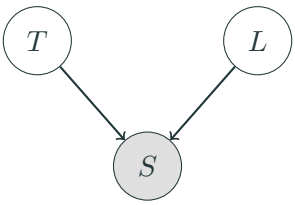
Bad control #1: something the treatment caused



D = thesis, M = first job, Y = earnings

- Maybe the thesis raises earnings partly by helping you land a research or consulting job.
- "Controlling for first job" then removes part of the effect you are trying to measure.
- Rule: don't control for outcomes of the treatment.
- Ask of every control: was it determined before or after the treatment?

Bad control #2: who ends up in your sample



T = talent, L = luck, S = startup gets funded

- Talent and luck are unrelated in the population.
- Among funded startups they are negatively related: a funded founder who isn't talented must have been lucky.
- Restricting your sample to S creates a correlation that isn't there. S is called a collider.
- Thesis version: studying only students who got Wall Street offers. Both the thesis and unobserved talent helped get them there.

A field guide to controls

The variable is…	Control for it?	Why
A common cause of D and Y	Yes	blocks a backdoor
A rough proxy for one	Usually	blocks it partially
A predictor of Y only (pre-treatment)	Yes	smaller standard errors
Caused by the treatment	No	removes part of the effect
Caused by both D and Y (or defines your sample)	No	creates bias

- "Add every control you have" is not a safe default.
- Further reading: Cinelli, Forney & Pearl (2022), "A Crash Course in Good and Bad Controls."

Selection on Observables

Matching: compare like with like

Thesis writers have better grades, take harder classes, have better recommendations...

- Matching: for each thesis writer, find a non-writer with the same GPA, major, test scores, etc. Use them as the counterfactual.
- Easy with a few variables. With many, you need to decide what "close" means (nearest neighbors, propensity scores, weighting).
- Always show a balance table: after matching, do the two groups look alike on what you observe?

Tools: `MatchIt` and `cobalt` in R; `teffects` in Stata.

The limits of matching

Matching assumes all differences between writers and non-writers are captured by what you observe.

- No selection on unobservables: ambition, curiosity, who your advisor knows…
- "Counterfactual you" = someone with your grades and test scores who chose not to write a thesis. Why did they choose that?

What helps:

- Use what you know about how treatment was assigned. Suppose only students *invited* to write a thesis can. Now compare invited students who wrote to invited students who didn't.
- Sensitivity analysis: how strong would an unobserved confounder have to be to explain away your result? (Oster 2019; Cinelli & Hazlett 2020, `sensemakr`.)

Usually the better answer is a research design that doesn't rely on having measured everything.

Where do counterfactuals come from?

Design	Counterfactual is…	You must believe…
Experiment / lottery	people who lost the coin flip	assignment really was random
Instrumental variables	people pushed away from treatment by something unrelated to Y	the push affects Y only through D
Event study	the same unit before the event	nothing else changed at that moment
Difference-in-differences	the change in an untreated group	trends would have been parallel
Synthetic control	a weighted mix of untreated units	the mix tracks the treated unit
Regression discontinuity	units just on the other side of a cutoff	nothing else jumps at the cutoff

Lotteries and Natural Experiments

Experiments someone else ran

- What about an actual experiment that someone else ran for other reasons?
- Or something that isn't an experiment but looks like one?
- We want variation in treatment that is as good as random.

Angrist, Pathak & Walters (2013): charter schools

Do charter schools improve test scores?

- Problem: families who apply to charters may care more about education. Comparing charter to public school students mixes the school effect with family effort.
- Solution: popular charters are over-subscribed and admit students by lottery.
- Compare lottery winners to losers. Both groups applied, so both care about education. Only luck differs.

Angrist, Pathak & Walters (2013)

TABLE 4—LOTTERY ESTIMATES OF CHARTER EFFECTS

	All charter schools		Urban charter schools		Nonurban charter schools	
	First stage	2SLS	First stage	2SLS	First stage	2SLS
Subject	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Middle school						
Math	1.02***	0.213***	1.03***	0.321***	1.01***	−0.123***
	(0.040)	(0.028)	(0.051)	(0.031)	(0.074)	(0.047)
N	16,543		11,941		4,602	
ELA	1.02***	0.075***	1.04***	0.146***	1.00***	−0.144***
	(0.040)	(0.025)	(0.051)	(0.028)	(0.074)	(0.039)
N	16,285		11,649		4,636	
Panel B. High school						
Math	0.565***	0.273***	0.508***	0.339***	1.13***	−0.020
	(0.085)	(0.071)	(0.090)	(0.077)	(0.197)	(0.071)
N	4,050		3,519		531	
ELA	0.565***	0.206***	0.508***	0.264***	1.14***	−0.046
	(0.086)	(0.060)	(0.090)	(0.067)	(0.196)	(0.059)
N	4,103		3,567		536	

Notes: This table reports 2SLS estimates of the effects of time spent in charter schools on test scores. The endogenous variable is years spent in charter schools and the instrument is a lottery offer dummy. Columns 1–2 show estimates for all schools, columns 3–4 show estimates for urban charter schools, and columns 5–6 show estimates for nonurban schools. The urban and nonurban estimates for a given subject come from a single regression with two endogenous variables, using urban and nonurban offers as instruments. All models control for race, sex, special education, limited English proficiency, subsidized lunch status, and a female by minority dummy. Year of birth, year of test, and risk set dummies are also included. Estimates for the middle school sample pool post-lottery outcomes for grades 4–8 and cluster by student identifier as well as school-grade-year. Estimates for the high school sample include only scores for tenth grade and cluster by school-grade-year.

How does it work?

Not every winner attends, and some losers get in later. So scale the effect of *winning*:

$$\text{effect of attending} = \frac{\text{score gap between winners and losers}}{\text{attendance gap between winners and losers}}$$

- If winning raises scores by 0.1 SD and raises attendance by 50 percentage points, then attending raises scores by $0.1/0.5 = 0.2$ SD.
- This is instrumental variables (IV). The lottery is the instrument.
- Two things must hold:
  1. The lottery actually changes who attends (relevance, which you can check).
  2. The lottery affects scores only through attendance (exclusion, which you must argue).

Whose effect does IV measure?

Think about four kinds of applicants:

Type	If they win	If they lose
Always-takers	attend	attend anyway (sibling, waitlist)
Never-takers	don't attend	don't attend
Compliers	attend	don't attend
Defiers (assume none)	don't attend	attend

- The lottery only changes behavior for compliers, so IV estimates the effect for compliers (a "local" average effect, LATE).
- A different instrument changes behavior for different people, so it can give a different (equally correct) answer.

Other lotteries and near-lotteries

- Angrist (1990): does military service affect later earnings?
  - Vietnam draft lottery numbers were random. White veterans earned roughly 15% less a decade later.
  - Exclusion worry: did low numbers also push men to stay in school to avoid the draft?
- Judge designs: cases are randomly assigned to judges, and some judges are harsher than others.
  - Dobbie, Goldin & Yang (2018): pretrial detention raises conviction rates and lowers later employment.
  - The same idea works for patent examiners, caseworkers, loan officers, and … thesis advisors.
- Weather: storms at sea cut the supply of fish at the Fulton Fish Market, which traces out demand (Graddy 2006).

In every case, ask: could the instrument affect the outcome some other way?

Sinkinson & Starc (2019): do drug ads work?

- Problem: firms set ad budgets as a share of revenue, and advertise where they expect ads to work. Ads and sales move together for reasons other than ads causing sales.
- Idea: during election season, political ads crowd out TV ad slots, so cholesterol-drug ads get displaced.
- Crowding out is worse in swing states and around primaries, and has nothing to do with how well drug ads work.
- That moves drug ads around for reasons unrelated to drug demand: an instrument.

Sinkinson & Starc (2019): political ads crowd out drug ads

Figure 4: National Pharmaceutical Ad Levels for Statins

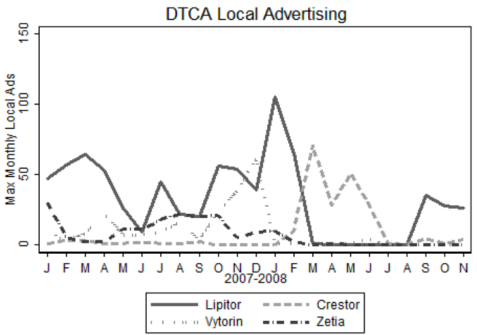
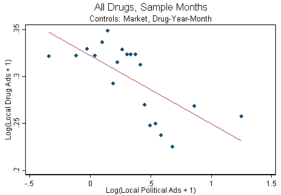


Figure 5: Political Ads Displace Local Drug Ads, Binned Scatter plot



Notes: The above plots bins of observations from July 2007 to November 2008 at the market-month level after residualizing by market and year-month fixed effects, and adding back the sample mean. Twenty bins are used. The fitted line is based on a regression of all underlying data, not only the binned values.

Sinkinson & Starc (2019): OLS vs. IV

Table 4: OLS Revenue Regressions for Advertised Drugs

	Dependent Variable: Log(Revenue per Insured)					
	This Month			Two Month Trailing Average		
Own Ads	0.0194***	0.0064***	0.0114***	0.0239***	0.0066***	0.0096***
	(0.0022)	(0.0020)	(0.0023)	(0.0021)	(0.0021)	(0.0023)
Rival Ads	0.0042*	0.0056**	0.0061**	0.0016	0.0036	0.0039
	(0.0025)	(0.0023)	(0.0026)	(0.0027)	(0.0026)	(0.0028)
Controls:						
Market FEs	X	X	X	X	X	X
Year FEs	X	X	X	X	X	X
Drug FEs	X	X	X	X	X	X
Drug FE*Time Trend		X	X		X	X
Drug FE*Time Trend^2			X			X
Drug-Year FEs	X			X		
N	11,551	11,551	11,551	11,550	11,550	11,550
R^2	0.842	0.847	0.848	0.843	0.847	0.848

Table 5: IV Revenue Regressions for Advertised Drugs

	Dependent Variable: Log(Revenue per Insured)					
	This Month			Two Month Trailing Average		
Own Ads	0.1559***	0.0808**	0.0923**	0.1252***	0.0764***	0.0734***
	(0.0251)	(0.0344)	(0.0373)	(0.0136)	(0.0258)	(0.0270)
Rival Ads	−0.1064***	−0.0492**	−0.0471**	−0.0966***	−0.0548***	−0.0425***
	(0.0179)	(0.0247)	(0.0261)	(0.0112)	(0.0212)	(0.0215)
Controls:						
Market FEs	X	X	X	X	X	X
Year FEs	X	X	X	X	X	X
Drug FEs	X	X	X	X	X	X
Drug FE*Time Trend		X	X		X	X
Drug FE*Time Trend^2			X			X
Drug-Year FEs	X			X		
N	11,551	11,551	11,551	11,550	11,550	11,550
R^2	0.755	0.819	0.822	0.788	0.824	0.832

- OLS (left): own ads barely matter. IV (right): own ads raise revenue several times more.
- With IV, rival ads turn negative: one drug's ads steal customers from the others.
- OLS was biased toward zero, so the naive comparison understated what ads do.

Before and After: Event Studies and Difference-in-Differences

Using the same unit before and after

On August 28, 2024, NVIDIA released one of the most anticipated earnings reports ever. What did it do to the stock price?

- Counterfactual: the stock's path just before the announcement, adjusted for what the overall market did that day.
- This is an event study. It works when the event is sharp, the window is short, and nothing else happens at the same time.

For most thesis questions (a policy, a merger, a new law) the window is months or years:

- Other things change too: the economy, other policies, trends already under way.
- We need someone who didn't get treated to tell us what "other things" did.

Combine the two counterfactuals: before vs. after and treated vs. untreated.

	Before	After	Change
Treated	A	B	$B - A$
Control	C	D	$D - C$
			$(B - A) - (D - C)$

- The control group's change tells us what would have happened to the treated group without treatment.
- The groups can start at different levels. They only need to be on the same trend.

The classic example: Card & Krueger (1994)

- April 1992: New Jersey raises its minimum wage from $4.25 to $5.05. Eastern Pennsylvania doesn't.
- Survey fast-food restaurants on both sides of the border before and after.
- Econ 101 predicts NJ employment falls relative to PA. It didn't.
- Decades of debate followed, about data, samples, and whether PA is the right control.

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE IN NEW JERSEY MINIMUM WAGE

	Stores by state			Stores in New Jersey[a]			Differences within NJ[b]	
Variable	PA (i)	NJ (ii)	Difference, NJ − PA (iii)	Wage = $4.25 (iv)	Wage = $4.26–$4.99 (v)	Wage ≥ $5.00 (vi)	Low–high (vii)	Midrange–high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	−2.69 (1.37)	−2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	−2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores[c]	−2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	−2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0[d]	−2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	−2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes: Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.

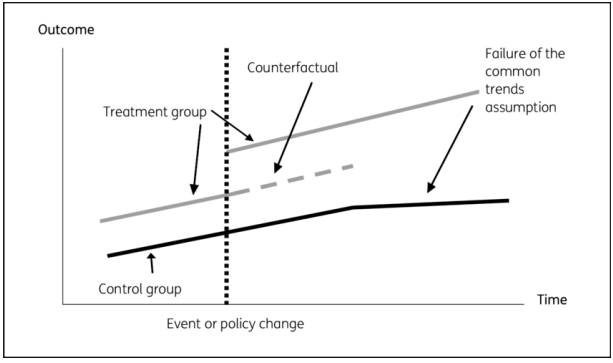
[a] Stores in New Jersey were classified by whether starting wage in wave 1 equals $4.25 per hour ($N = 101$), is between $4.26 and $4.99 per hour ($N = 140$), or is $5.00 per hour or higher ($N = 73$).

[b] Difference in employment between low-wage ($4.25 per hour) and high-wage (≥ $5.00 per hour) stores; and difference in employment between midrange ($4.26–$4.99 per hour) and high-wage stores.

[c] Subset of stores with available employment data in wave 1 and wave 2.

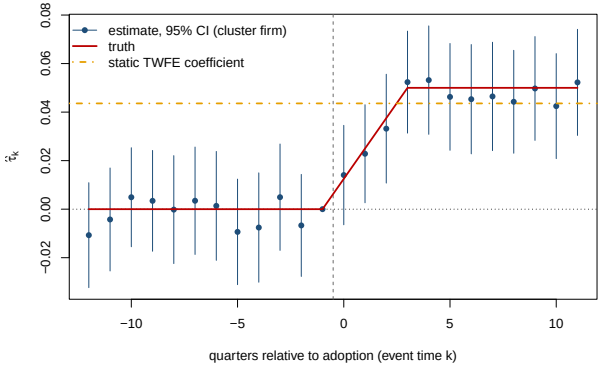
[d] In this row only, wave-2 employment at four temporarily closed stores is set to 0. Employment changes are based on the subset of stores with available employment data in wave 1 and wave 2.

The key assumption: parallel trends



- Without treatment, the treated group would have followed the same trend as the control group.
- It is about the post-period counterfactual, which we never see. You cannot test it, only make it plausible.
- Choose controls that face the same shocks: neighboring states, similar firms, the same market.

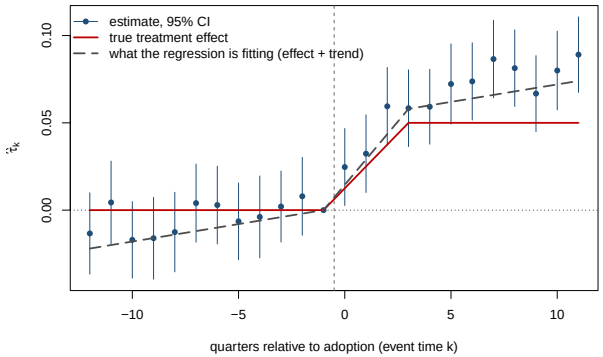
The modern way to show a DiD: the event-study plot



- One DiD estimate per period, relative to the period just before treatment ($k = -1$, zero by construction).
- Left of the line: were treated units already diverging? Should be flat.
- Right of the line: how the effect evolves. Does it ramp up? Fade out?
- A single DiD number averages over this path. Show the path.

Simulated data: 400 firms adopt a pricing tool in quarter 13; true effect in red.

Flat pre-trends do not prove parallel trends



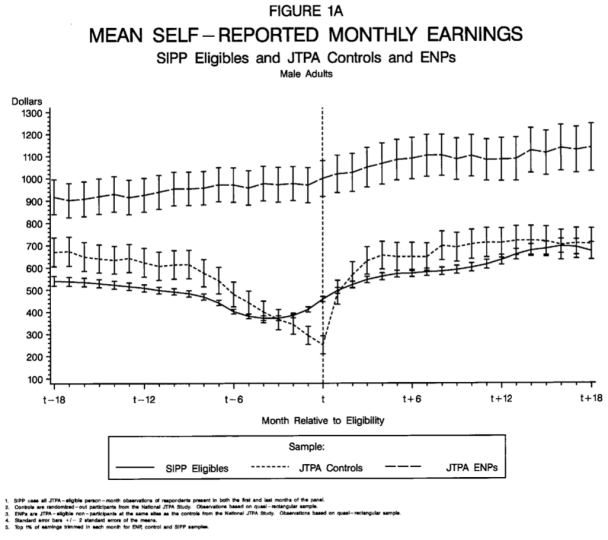
- Same data plus a small extra trend for treated firms. No pre-period estimate is significant, but every post-period estimate is too big.
- "The pre-trends are insignificant" is weak evidence. A noisy test passes easily (Roth 2022).
- Better: say how big a trend your data can rule out, and how much it could move your answer (Rambachan & Roth 2023, `HonestDiD`).

What breaks parallel trends?

1. Something else changed at the same time. If NJ raised the minimum wage because a new governor arrived with a whole package of policies, NJ and PA diverge for other reasons too, even if they tracked each other before.
2. Selection on a temporary dip. People join job training after a bad year. Their earnings would have bounced back anyway (next slide).
3. Anticipation. Firms react to a law when it is announced, not when it takes effect.
4. Composition. Treatment changes who is in the sample: restaurants close in NJ and reopen in PA, or workers move.

For each, ask: why did this unit get treated at this time?

The “Ashenfelter dip” (Heckman & Smith 1999)



- Earnings of job-training participants drop just before they enroll.
- Some of the rebound afterwards would have happened anyway (regression to the mean).
- DiD starting from the dip overstates the program’s effect.

Staggered adoption: when treatment arrives at different times

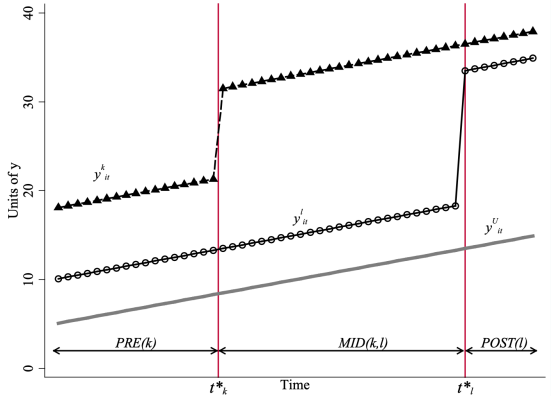
Most policies roll out gradually: states adopt a law in different years, firms adopt a technology one by one.

- Example: U.S. states adopted unilateral (no-fault) divorce between 1969 and 1985. Did it change female suicide rates? (Stevenson & Wolfers 2006)
- Standard approach until about 2020: regress the outcome on a treatment dummy with state and year fixed effects ("two-way fixed effects").
- It seems strictly better than one treated state. It isn't automatically.

Since 2020, a large literature (Goodman-Bacon; Callaway & Sant'Anna; Sun & Abraham; de Chaisemartin & D'Haultfœuille; Borusyak, Jaravel & Spiess) shows what that regression actually compares.

Three groups: early, late, never treated

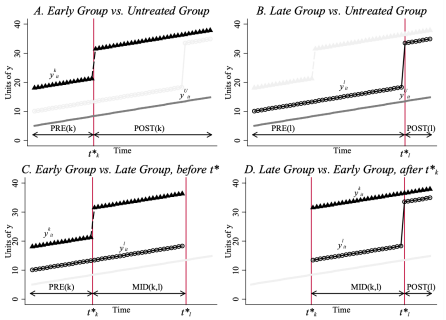
Figure 1. Difference-in-Differences with Variation in Treatment Timing: Three Groups



Notes: The figure plots outcomes in three groups: a control group, U, which is never treated; an early treatment group, E, which receives a binary treatment at $t_k^* = \frac{34}{100}T$; and a late treatment group, ℓ, which receives the binary treatment at $t_\ell^* = \frac{85}{100}T$. The x-axis notes the three sub-periods: the pre-period for group k, $[1, t_k^* - 1]$, denoted by $PRE(k)$; the middle period when group k is treated and group ℓ is not, $[t_k^*, t_\ell^* - 1]$, denoted by $MID(k, \ell)$; and the post-period for group ℓ, $[t_\ell^*, T]$, denoted by $POST(\ell)$. I set the treatment effect to 10 in group k and 15 in group ℓ.

What the regression secretly compares (Goodman-Bacon 2021)

Figure 2. The Four Simple (2x2) Difference-in-Differences Estimates from the Three Group Case



Notes: The figure plots the groups and time periods that generate the four simple 2x2 difference-in-difference estimates in the case with an early treatment group, a late treatment group, and an untreated group from Figure 1. Each panel plots the data structure for one 2x2 DD. Panel A compares early treated units to untreated units ($\hat{\beta}^{2DD}_{kU}$); panel B compares late treated units to untreated units ($\hat{\beta}^{2DD}_{lU}$); panel C compares early treated units to late treated units during the late group's pre-period ($\hat{\beta}^{2DD,k}_{kl}$); panel D compares late treated units to early treated units during the early group's post-period ($\hat{\beta}^{2DD,l}_{kl}$). The treatment times mean that $\overline{D}_k = 0.67$ and $\overline{D}_l = 0.16$, so with equal group sizes, the decomposition weights on the 2x2 estimate from each panel are 0.365 for panel A, 0.222 for panel B, 0.278 for panel C, and 0.135 for panel D.

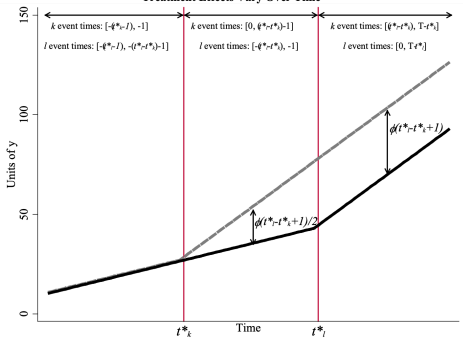
The regression averages every possible 2×2 DiD:

A. Early vs. never treated — fine
B. Late vs. never treated — fine
C. Early vs. late, before the late group is treated — fine
D. Late vs. early, *after* the early group is treated — problem

In D, an already-treated group serves as the control.

Why comparison D is a problem

Figure 3. Difference-in-Differences Estimates with Variation in Timing Are Biased When Treatment Effects Vary Over Time



Notes: The figure plots a stylized example of a timing-only DD set up with a treatment effect that is a trend-break rather than a level shift (see Meer and West 2013). Following section II.A.ii, the trend-break effect equals $\phi \cdot (t - t^* + 1)$. The top of the figure notes which event-times lie in the $PRE(k)$, $MID(k, \ell)$, and $POST(\ell)$ periods for each unit. The figure also notes the average difference between groups in each of these periods. In the $MID(k, \ell)$ period, outcomes differ by $\frac{\phi}{2}(t_\ell^* - t_k^* + 1)$ on average. In the $POST(\ell)$ period, however, outcomes had already been growing in the early group for $t_\ell^* - t_k^*$ periods, and so they differ by $\phi(t_\ell^* - t_k^* + 1)$ on average. The 2x2 DD that compares the later group to the earlier group is biased and, in the linear trend-break case, weakly negative despite a positive and growing treatment effect.

- If the effect grows over time, the early group is still changing because of *its own* treatment.
- Using it as a control subtracts part of the treatment effect.
- The overall estimate can be badly biased, and even have the wrong sign, though every unit's effect is positive.

The fix: only make clean comparisons

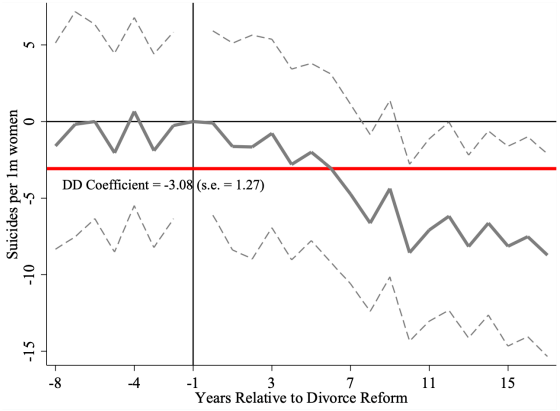
The idea behind the modern estimators is simple:

1. For each cohort (units treated in the same year), estimate a separate DiD using only never-treated or not-yet-treated units as controls.
2. Then average those clean estimates yourself: by cohort, by calendar year, or by time since treatment (an event-study plot).

- R: `did` (Callaway & Sant'Anna), `fixest::sunab` (Sun & Abraham), `didimputation`. Stata: `csdid`, `did_imputation`, `jwdid`.
- If everyone is treated at the same time, the classic regression is fine. If timing varies, use one of these.

Event study done right: no-fault divorce

Figure 5. Event-Study and Difference-in-Differences Estimates of the Effect of No-Fault Divorce on Female Suicide: Replication of Stevenson and Wolfers (2006)



Notes: The figure plots event-study estimates from the two-way fixed effects regression equation on page 276 and plotted in figure 1 of Stevenson and Wolfers (2006), along with the DD coefficient. The specification does not include other controls and does not weight by population. Standard errors are robust to heteroskedasticity.

A DiD checklist for your thesis

1. Who is treated, who is the control, and when does treatment start?
2. Why did treated units get treated at that time? Is that reason related to the outcome?
3. Plot the raw averages for each group over time before running any regression.
4. Show the event-study plot. Discuss how big a pre-trend you could have missed.
5. If timing is staggered, use a modern estimator, not plain two-way fixed effects.
6. Cluster standard errors at the level treatment is assigned (states, not people). With only a few treated states, be humble about precision.

One Treated Unit: Synthetic Control

Abadie, Diamond & Hainmueller (2010): California's Prop 99

In 1988 California passed Proposition 99:

- Raised the cigarette tax by 25 cents per pack
- Funded anti-smoking education and ads
- Followed by local indoor smoking bans

What was the effect on cigarette sales per person?

- Only one treated state, and sales were already falling everywhere.
- Which state is a good control for California? Nevada? Oregon? None of them is very good on its own.

The idea: build a “synthetic California”

- Take a weighted average of other states that matches California’s cigarette sales every year before 1988, and its income, prices, and demographics.
- Weights are non-negative and add to one. The data choose them, not you.
- After 1988, the gap between real and synthetic California is the estimated effect.

- Like matching: build a control that looks like the treated unit.
- Like DiD: compare paths before and after.
- Credibility comes from how well the pre-period fits, so show it.

Synthetic California

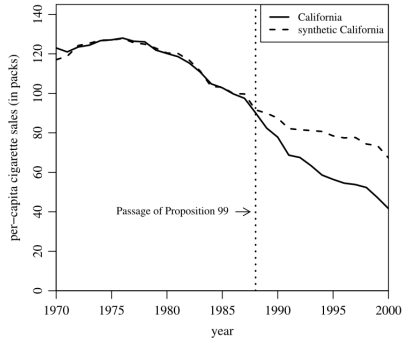
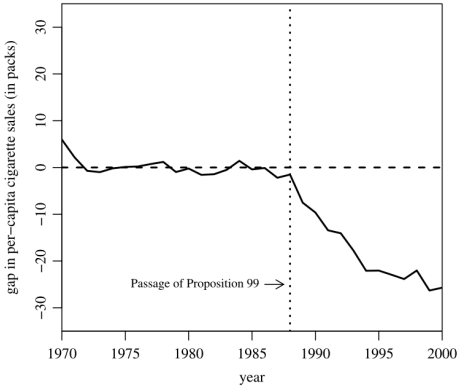


Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.



Synthetic California is mostly Utah, Nevada, Montana, Colorado, and Connecticut. Left: real vs. synthetic sales. Right: the gap.

Is California unusual? Placebo tests

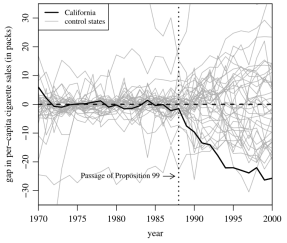


Figure 4. Per-capita cigarette sales gaps in California and placebo gaps in all 38 control states.

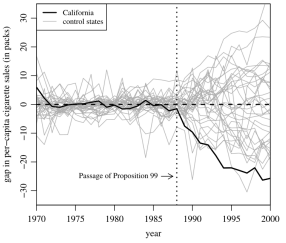


Figure 5. Per-capita cigarette sales gaps in California and placebo gaps in 34 control states (discards states with pre-Proposition 99 MSPE twenty times higher than California's).

- Pretend each other state passed Prop 99 in 1988 and build a synthetic control for it.
- If California's gap looks like the placebo gaps, we've learned nothing.
- Here California stands out. This comparison is the inference.

DiD, synthetic control, or synthetic DiD?

All three are weighted comparisons of treated and control units before and after:

- Many treated units, similar trends: difference-in-differences.
- One or a few treated units that a mix of controls can mimic: synthetic control.
- A treated unit whose level is unusual but whose trend is not: synthetic DiD (Arkhangelsky et al. 2021). It weights controls to match the trend, then differences out the level gap.
- A treated unit with an extreme trend (fastest-growing city, biggest firm): none of them works. Say so.

R: `Synth`, `tidysynth`, `synthdid`.

Cutoffs: Regression Discontinuity

Rules with cutoffs are everywhere

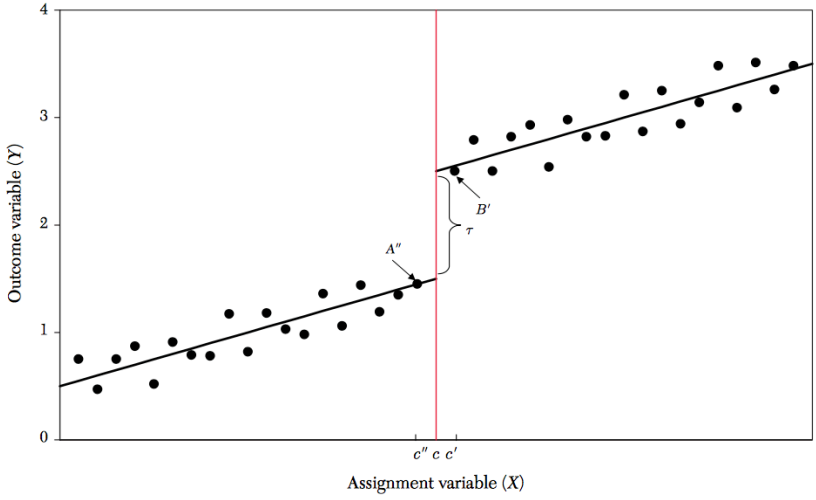
Suppose Stern's honors program required a GPA of at least 3.5 to write a thesis.

- Students at 3.49 and 3.51 are nearly identical: same ability, same ambition, give or take a lucky grade.
- But only one of them can write a thesis.
- Compare outcomes just above vs. just below the cutoff: that is close to a lottery.

This is regression discontinuity (RD). The variable that determines treatment (GPA) is the running variable.

- Cutoffs show up in test-score admissions, vote shares, age limits, income eligibility, firm-size rules, ratings...

RD in a picture



What has to be true

- Nothing else jumps at the cutoff. If 3.5 also triggers Latin honors or a recruiting list, you can't separate the thesis effect from those.
- People can't precisely sort around the cutoff.
  - If students can lobby professors for a grade bump to 3.5, the ones just above differ from the ones just below.
  - Examples: under-reporting income to qualify for a program, drug quantities charged just below thresholds.
- You learn the effect at the cutoff. Effects for 3.0 or 3.9 students may differ.
- If crossing the cutoff only raises the *chance* of treatment, you have a fuzzy RD: it works like IV, and the cutoff is the instrument.

Lee (2008): do incumbents have an advantage?

- Incumbents win re-election far more often. Does holding office help, or are good politicians just good?
- Running variable: the Democrat's vote margin in the previous House election.
- Compare districts where Democrats barely won to districts where they barely lost.
- Barely winning vs. barely losing a close election is close to a coin flip.

Lee (2008)

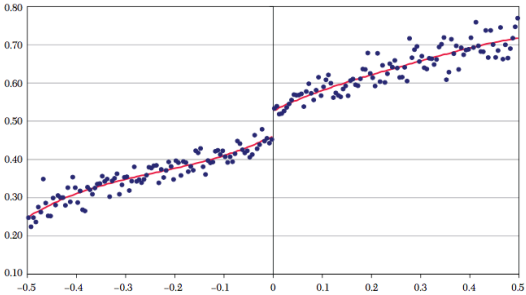


Figure 8. Share of Vote in Next Election, Bandwidth of 0.005 (200 bins)

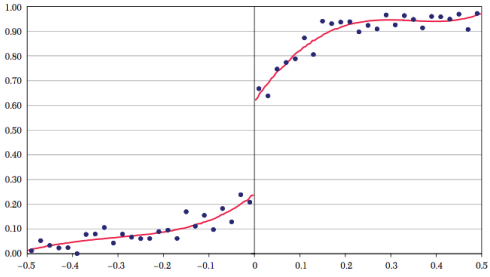
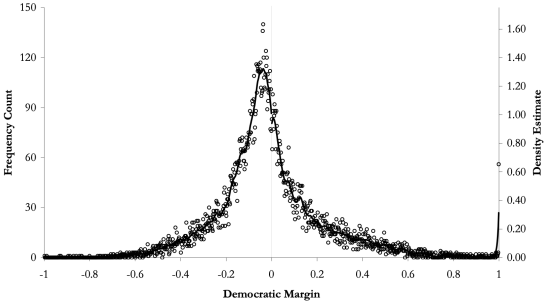


Figure 9. Winning the Next Election, Bandwidth of 0.02 (50 bins)

Barely winning raises the chance of winning next time by about 40 percentage points.

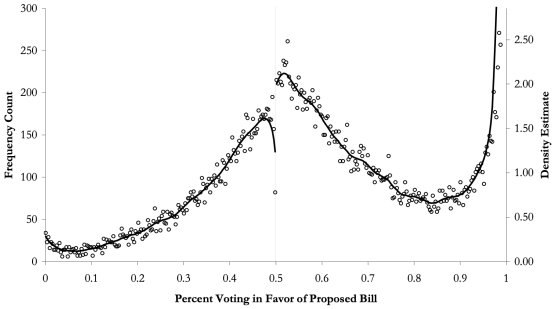
Checking for sorting (McCrary 2008)

Figure 4. Democratic Vote Share Relative to Cutoff: Popular Elections to the House of Representatives, 1900-1990



House elections: smooth density

Figure 5. Percent Voting Yeay: Roll Call Votes, U.S. House of Representatives, 1857-2004



Roll-call votes: bunching just above 50%

If the number of units jumps at the cutoff, people are sorting and the design is in trouble.

What every RD paper shows

1. A plot of treatment against the running variable: the jump in treatment.
2. A plot of the outcome: the jump you are claiming.
3. Plots of pre-determined characteristics: these should *not* jump.
4. The density of the running variable: no bunching.
5. Estimates using a local linear fit near the cutoff, robust to the choice of bandwidth.

- Avoid high-order polynomials fit to the whole range (Gelman & Imbens 2019).
- Tools: `rdrobust`, `rddensity`, `rdplot` (Cattaneo, Idrobo & Titiunik) in R, Stata, and Python.

More cutoffs

- Yelp ratings (Luca): a restaurant's average gets rounded to the nearest half-star. Compare 3.74 to 3.76 to measure the effect of an extra half star on revenue.
- Class size (Angrist & Lavy 1999): Israeli schools split classes when enrollment passes 40, so class size jumps at 41, 81, ...
- Elite exam schools (Abdulkadiroğlu, Angrist & Pathak 2014): students just above vs. just below the admission score in Boston and New York do about the same later. The "elite" effect is mostly selection.

Final Thoughts

Always be asking

- What am I trying to measure, and for whom?
- What is my counterfactual?
- Why do treated and untreated units differ? What is my selection problem?
- What plausibly random variation can I use?
- What must be true for my design to work, and what evidence can I show for it?
- Am I controlling for something the treatment caused, or choosing my sample on an outcome?

Don't look for your keys only where the light is. But you could do a lot worse than following one of the scripts above.

Where to read more

- Angrist & Pischke, *Mastering 'Metrics*: the most readable introduction, written for undergraduates.
- Huntington-Klein, *The Effect* (theeffectbook.net): free, with code in R, Stata, and Python.
- Cunningham, *Causal Inference: The Mixtape* (mixtape.scunning.com): free, with code.
- Roth, Sant'Anna, Bilinski & Poe (2023), "What's Trending in Difference-in-Differences?": a readable survey of the new DiD methods.